

Working with Files in Linux: tar, gzip, gunzip, truncate & split

tar

What it does

- ◆ Combines multiple files/folders into a single archive (like packing into a box).
- ◆ Can be optionally compressed with gzip or bzip2.

Basic usage

```
tar -cvf archive.tar file1 file2
```

```
tar -xvf archive.tar
```

```
tar -czvf archive.tar.gz folder/
```

```
tar -xzvf archive.tar.gz
```

✓ Important Points

- ◆ -c → create archive
- ◆ -x → extract archive
- ◆ -v → verbose, shows files being processed
- ◆ -f → specify archive filename
- ◆ -z → gzip compression
- ◆ -j → bzip2 compression
- ◆ -C → change directory while extracting

gzip

What it does

- ◆ Compresses a single file to reduce size. Produces .gz.
- ◆ Think of it as **squeezing a file**.

Basic usage

```
gzip file.txt # compress → file.txt.gz
```

```
gzip -d file.txt.gz # decompress → restores original file
```

✓ Important Points

- ◆ Works on single files (use tar for multiple files).
- ◆ Compression is faster than zip in most cases.

gunzip

What it does

- ◆ Decompress .gz files (reverse of gzip).

Usage

```
gunzip file.gz
```

✓ Important Points

- ◆ Original file is restored automatically.
- ◆ Equivalent to gzip -d file.gz.

truncate

What it does

- ◆ Shrinks or extends a file to a specified size without editing manually.
- ◆ Think of it as **resizing a file** instantly.

Basic usage

```
truncate -s 1K file.txt # shrink file to 1 KB
```

```
truncate -s +1K file.txt      # extend file by 1 KB (filled with zeros)
```

✓ Important Points

- ◆ Useful for testing large files.
- ◆ Can clear log files quickly.

split

What it does

- ◆ Splits a large file into smaller chunks.
- ◆ Think of it as **cutting a big cake into slices**.

Basic usage

```
split -b 1M bigfile.txt part_      # split into 1 MB chunks
split -l 1000 log.txt log_         # split into 1000 lines per
file
cat part_* > bigfile.txt           # merge back
```

✓ Important Points

- ◆ -b → size in bytes (K, M, G allowed)
- ◆ -l → number of lines per file
- ◆ Useful for transferring or processing large files

Quick Summary Table

Command	Purpose
tar -cvf	Create archive
tar -xvf	Extract archive
tar -czvf	Create archive + gzip compression
tar -xzvf	Extract gzip archive
gzip	Compress a file
gunzip / gzip -d	Decompress a .gz file
truncate	Shrink or extend file
split -b	Split file by bytes
split -l	Split file by lines
cat part_*	Merge split files back